

ePiE

exposure to Pharmaceuticals in the Environment

ePiE technical model description

Version 1.2

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List of abbreviations

API	Active Pharmaceutical Ingredient
WWTP	Wastewater Treatment Plant
WP	Work Package
iPiE	Intelligence-led Assessment of Pharmaceuticals in the Environment (project)
ePiE	Exposure to Pharmaceuticals in the Environment (model)
QSAR	Quantitative Structure-Activity Relationship
QSPR	Quantitative Structure-Property Relationship
QSBR	Quantitative Structure-Biodegradation Relationship
DoW	Description of Work
DDD	Defined Daily Dosages
p.e.	population equivalents
UWWTD	Urban Waste Water Treatment Directive
EEA	European Environmental Agency
CRU TS	Climatic Research Unit Time-Series
NCEP/NCAR	National Centers for Environmental Prediction / National Center for Atmospheric Research
HydroSHEDS	Hydrological data and maps based on SHuttle Elevation Derivatives at multiple Scales
FLO1K	Streamflow dataset at 1 km resolution
DEM	Digital Elevation Map

1. Introduction

Human active pharmaceutical ingredients (APIs) are used in large quantities, and can partly end up in the environment because they are generally not completely broken down during passage through the human body and wastewater treatment plants (WWTPs). Pharmaceutical residues are mainly transported into the environment via the effluent of WWTPs. Although concentrations reported are generally low, i.e. in the ng/L to µg/L range (aus der Beek et al., 2015; Fatta-Kassinos et al., 2011; Kümmeler, 2010), adverse ecological effects caused by some pharmaceuticals are plausible considering their specific modes of action and high potency. A crucial step in environmental risk assessment of chemicals is the estimation of their environmental exposure potential. Because there are currently thousands of different pharmaceuticals in use, however, monitoring all of them everywhere in Europe all of the time is practically impossible. The need thus exists for tools and methods that enable a reliable location-specific prediction of environmental concentrations. Moreover, such tools and methods will not only facilitate directed monitoring and location-specific environmental risk assessment of existing APIs, but could also support environmental considerations during the development process of new APIs. This implies that input requirements should correspond to the simple physicochemical properties generally available during development, or that more advanced input data can be derived using QSAR/QSPR/QSBR approaches.

Oldenkamp et al. (2013) originally developed a decision support tool to enable European-wide and spatially explicit prioritisation of APIs. It provides regionalized estimates of the relative impacts of pharmaceuticals throughout Europe, for both the aquatic environment and human health. It is based on the four consecutive steps of emission, fate, exposure and effect. As a first step, it calculates emissions into wastewater as the sum of patient non-compliance and subsequent disposal via flushing, and of actual consumption and excretion as parent compound via urine and faeces. Pharmaceutical residues in wastewater are estimated at the level of individual Member States, before they are divided over individual agglomerations (i.e. urban human settlements comprising of built-up areas), based on their population size. Generally, this wastewater is discharged into the surface water after passage through a WWTP. However, part of the wastewater might also be discharged into the surface water directly, depending on the local level of WWTP-connectivity. Indirect emissions (i.e. after passage through WWTP) are calculated at the level of the individual WWTPs, and depend on WWTP design and API-specific characteristics. Furthermore, the model also estimates emissions of APIs to agricultural soils, which depend on pharmaceutical levels in secondary sewage sludge and Member State specific sludge disposal practices. The model subsequently aggregates emissions to surface water and agricultural soils at the level of 100 km * 100 km environmental grid cells, spatially parameterized with data from Pistocchi et al. (2006). This is done to enable multimedia fate calculations with the level III (steady-state) multimedia fate model SimpleBox 3.0 (e.g. Hollander et al., 2009). Consequently, the calculations result in long-term (yearly) average steady-state surface water and soil concentrations.

As described in the iPiE (Intelligence-led Assessment of Pharmaceuticals in the Environment) project description of work (DoW), the emission and environmental fate modules of the model by Oldenkamp et al. (2013) were originally envisioned to form the basis of the ePiE (exposure to Pharmaceuticals in the Environment) model to be developed within WP3 of the project. However, during teleconference meetings it became clear that the ability of hot-spot identification was considered most useful as a potential application of the ePiE modelling

framework. This requires a different approach to environmental fate modelling than the multimedia fate approach implemented in the original model by Oldenkamp et al. (2013), because that does not provide the spatial resolution needed to pinpoint specific concentrations at specific locations. However, since APIs follow a relatively straightforward emission pathway into the environment (i.e., WWTPs mainly function as point source emissions), such multimedia fate calculations are not required for concentration estimations in surface waters.

This report describes the technical background of the ePiE model (Oldenkamp et al., 2018); a model that predicts concentrations of pharmaceuticals across Europe based on national consumption data. ePiE predicts environmental concentrations in surface waters (rivers and lakes) and their upper sediment layers. The ePiE model uses state of the art spatial artificial neural network models to estimate local year-specific streamflow metrics (annual average, minimum and maximum monthly flow). Combined with spatially explicit information on the location and characteristics of WWTPs throughout Europe, this enables concentration predictions for surface waters at specific coordinates or at a specific distance downstream of the point of emission. Moreover, because it applies directed calculations through the river network, ePiE incorporates the contribution of upstream sources in the assessment of individual WWTPs. This approach provides an advantage over traditional process-based hydrological models, which might be overly time-consuming and data-intensive and often do not provide the ability to pinpoint locations of interest.

There are currently two versions of the ePiE model, i.e. an R version and a version in the iPiEsys software. Originally, the ePiE model was developed in R and this version was later implemented in iPiEsys. All European basins, with the exception of the Danube and the Dnjepr basins, are currently available in the R version of ePiE, although only 8-10 basins have actually been tested. Because of computational limitations, the ePiE version in iPiEsys contains only 8 river basins, i.e. the Odense basin in Denmark, the Wear, Ouse and Dove basins in the UK, the Rhine basin in Northern-Western Europe, and the Santiago, Turia and Guadalquivir basins in Spain.

The technical description of ePiE starts with Chapter 2 **Fout! Verwijzingsbron niet gevonden.**, in which equations are described that estimate yearly emissions of APIs from the WWTPs to surface waters, based on different treatment techniques applied and the removal of APIs related to them. These emissions form the starting point for Chapter 3, in which the estimations of API concentrations in river water, lakes and reservoirs are defined; in their water column as well as the upper sediment layers.

The partitioning of APIs between a range of media and water is incorporated at different points throughout the ePiE model. Equations that present estimations of solids-water and air-water partitioning coefficients are described in Chapter 4. The ePiE model can account for five environmental dissipation processes: biodegradation, photolysis, hydrolysis, volatilisation, and sedimentation. These dissipation processes are explained in detail in Chapter 5.

2. Emission estimations

The ePiE model follows the exposure assessment chain combining information at different levels of spatial resolution with physicochemical properties of the API assessed. As a starting point, it requires the input of yearly consumption data (kg/year) in the countries that the river basin of interest covers. For APIs that are formed as an active metabolite of another API, i.e., their prodrug, consumption data for that prodrug are also required. If data are only available in another form (i.e. the number of Defined Daily Dosages (DDD) or packages sold), users themselves need to translate these into the total national amount based on demographics. The ePiE model does not allow for the input of regional consumption data, since administrative/political boundaries and hydrological boundaries and sewerage service areas often do not properly align.

API loads entering the wastewater are then estimated using information on human metabolism. After oral consumption, APIs can be absorbed to different extents from the intestinal tract, entering the blood circulation. The non-absorbed fraction might end up in the faeces and as such can be egested into the wastewater. The absorbed fraction might partly be metabolized in the body, and partly be excreted via urine, or via bile into the faeces. To estimate the load entering the wastewater, the consumption thus needs to be adjusted by accounting for the fraction of the administered dose excreted as parent compound (and reversible conjugates) via urine, and the fraction of the administered dose excreted/egested as parent compound via faeces. A combined summation of these fractions is required as user input in ePiE. In addition, ePiE requires the fraction of any prodrug dosage that is excreted/egested as the API of interest (or as its reversible conjugate). Currently, ePiE does not specifically accommodate the estimation of dermally administered APIs, of which a part might enter wastewater directly after application. However, the user might incorporate knowledge on this fraction in the overall fraction excreted/egested unchanged.

The yearly loads present in the wastewater are distributed over agglomerations in the relevant country based on their respective generated wastewater loads (expressed in population equivalents; p.e.) as a proxy of their relative population size (Chapter 2.1). Per WWTP, the API mass load in the influent is subsequently calculated from the level of connectivity of all agglomerations connected to it. Direct emissions from agglomerations are included for those that do not have full connectivity to WWTPs (Chapter 2.1). Finally, yearly emissions are estimated from the WWTPs, based on the treatment techniques applied and the removal of APIs related to them. *Equation 1* presents the direct emission estimation to surface waters from individual agglomerations with incomplete WWTP connectivity; *Equation 2* presents the emission estimation to surface waters from individual WWTPs.

$$E_{w,agg} = (M \cdot f_{pc} + M_{pd} \cdot f_{met}) \cdot \frac{V_{ww,agg}(1-f_{conn,agg})}{V_{ww,cnt}} \quad \text{Equation 1}$$

- $E_{w,agg}$ is the emission to surface water from an agglomeration (kg/year);
- M is the yearly consumption of API of interest in the relevant country (kg/year) (user input);
- M_{pd} is the yearly consumption of the prodrug of interest in the relevant country (kg/year) (user input);
- f_{pc} is the combined fraction of the administered dose excreted/egested unchanged or as reversible conjugates via urine/faeces (-) (user input);

- f_{met} is the fraction of prodrug metabolized to the API of interest, and subsequently excreted/egested via urine and faeces (-) (user input);
- $V_{ww,agg}$ is the volume of wastewater generated by the agglomeration (p.e.) (Chapter 2.1);
- $V_{ww,cnt}$ is the total volume of wastewater generated in the relevant country, calculated by summing the volumes of all individual WWTPs in that country (p.e.) (Chapter 2.1);
- $f_{conn,agg}$ is the level of WWTP connectivity of the agglomeration (-) (Chapter 2.1).

$$E_{w,wwtp} = (M \cdot f_{pc} + M_{pd} \cdot f_{met}) \cdot \frac{\sum_{j=1}^n (V_{ww,agg,j} \cdot f_{conn,agg,j} \cdot f_{wwtp,agg,j})}{V_{ww,cnt}} \cdot (1 - f_{rem})$$

Equation 2

- $E_{w,wwtp}$ is the emission to surface water from a WWTP (kg/year);
- n is the number of agglomerations (partially) connected to the WWTP (-) (Chapter 2.1);
- $f_{wwtp,agg,j}$ is the fraction of agglomeration j connected to the WWTP (-) (Chapter 2.1);
- f_{rem} is the API-specific removal efficiency of the WWTP (-) (Chapter 2.2).

Information on the spatial distribution and characteristics of agglomerations and WWTPs in Europe was derived from the UWWTD (Urban Waste Water Treatment Directive) database (EEA, 2015). A description of this database and the curation exercise that was performed to make it suitable for calculations in ePiE is provided in Chapter 2.1. Additionally, Chapter 2.2 contains a description of the calculation of the fate of APIs during their transport through WWTPs. Finally, Chapter 2.3 describes the allocation of specific locations of emissions to the individual agglomerations and WWTPs, which form the starting point for the estimation of environmental concentrations (Chapter 0).

2.1. European wastewater treatment plants and agglomerations

Version 5 of the UWWTD database (uploaded 19/02/2015, archived 14/12/2017) from the European Environmental Agency was used (EEA, 2015). This database contains spatial information on the location and characteristics of 27,695 European agglomerations, and 30,043 urban WWTPs and wastewater collection systems. Agglomerations with a generated load below 2,000 p.e. are not included in the database, since the directive does not require reporting on them.

The agglomeration characteristics reported in UWWTD database relevant for the ePiE model are their location (longitude and latitude coordinates), their generated load (p.e.), and the fraction connected to WWTP (-). Similarly, the UWWTD database contains for each WWTP the location (longitude and latitude coordinates), the load entering (p.e.), as well as its design capacity (p.e.). It must be noted that some WWTPs receive a combination of urban and industrial wastewater, of which the database only reports the combined load. Consequently, API loads towards such WWTPs might deviate from the estimations. In addition, each WWTP is assigned an identification number related to the agglomeration connected to it, and the level of treatment is indicated (primary, secondary, or tertiary treatment). For those WWTPs applying tertiary treatment, this is further specified as the type of advanced treatment, distinguishing between N-removal, P-removal, chlorination, ozonation, UV-treatment, and sand filtration.

Curation of the UWWTD database: missing and erroneous coordinates

Although the database is extensive, it does contain some erroneous, missing, or ambiguous entries. Therefore, a curation of the database was performed. First, both agglomerations and WWTPs with missing coordinates or mixed up latitude and longitude were identified. This was the case for 1.2% of the agglomerations in the database. A few of these (16 records) had mixed up latitude and longitude, which was corrected. For another 13 agglomerations no coordinates were reported because their generated load was below 2,000 p.e.. These were excluded from further calculations. The remaining 292 agglomerations without correct coordinates (1.1% of the initial set) were also excluded from further calculations. As a result, coordinates were available for 27,381 agglomerations in the database. From these, 799 were indicated in the database as 'inactive', generally due to merging of agglomerations into one record that were separately included in a previous version of the database, or because agglomerations dropped below the 2,000 p.e. size threshold for reporting. These agglomerations were also excluded from further calculations, resulting in a total of 26,582 agglomerations used in further calculations.

Of the wastewater treatment plants in the database, 9.4% had missing or erroneous coordinates. Of these records, a few (11 records) had mixed up latitude and longitude, which was corrected. The exclusion of all WWTPs indicated in the database as being inactive reduced the database to 27,432 WWTPs, and increased the percentage with correct coordinates to 94.2%. Of the remaining 5.8% of WWTP-records without correct coordinates (1,596 records), 784 were assigned coordinates of the agglomeration to which they were linked via the agglomeration identification number. Although this likely has led to a loss of accuracy in the estimation of the emission point for these WWTPs (location of WWTP now equals location of agglomeration linked to it), it was considered preferable over their exclusion. This resulted in a database with 26,620 WWTP-records with coordinates. The remaining 818 WWTP-records were removed from the database and excluded from further calculations. These were mainly Southern Italian wastewater collecting systems without treatment connected to them. This resulted in 26,614 WWTP-records with coordinates in the database.

Curation of the UWWTD database: linking WWTPs to agglomerations

In principle, each WWTP included in the database is assigned an agglomeration identification number, ensuring that it is linked to an agglomeration in the database. In practice, however, only 12,765 of the 26,614 WWTP records in the database (after curation for missing and erroneous coordinates) had been assigned such an agglomeration identification number, leaving 13,849 without. In order to make a WWTP-based approach possible, the agglomerations connected to these WWTPs had to be deduced based on similarities in names, codes, coordinates, or loads generated. By doing this, we managed to increase the number of WWTPs with an agglomeration ID attached to it to 25,697. The remaining WWTPs were mainly located in Southern Italy. The 1,735 WWTP records that were excluded from the total database (818 due to missing/erroneous coordinates; 917 due to lack of agglomeration ID), represent ~3% of the total wastewater load entering all WWTPs in Europe. These WWTPs were included in the spatial distribution of the total consumption over agglomerations and WWTPs, but were excluded from further concentration calculations.

After linking each WWTP to an agglomeration, 3,961 agglomerations were not yet connected to any WWTP record. For 2,762 of these agglomerations, this is due to their 0% connectivity to wastewater collection and WWTPs. Instead, their wastewater might be (partly) addressed

in independent appropriate systems (IAS), and might be (partly) discharged untreated. While the database does distinguish between these two options, we conservatively assumed direct discharge without treatment for all wastewater not directed towards a WWTP. From the remaining 1,199 agglomerations not connected to any WWTP, 562 could be linked to a WWTP serving multiple agglomerations (to which another agglomeration had already been linked). The final 637 agglomerations could not be linked to any WWTP. These were located in Southern Italy (428) and Croatia (209). For the Croatian records, this was due to the lack of information on their WWTP connectivity (i.e., all Croatian agglomerations reported zero connectivity). Without knowing their WWTP connectivity, it was not possible to determine whether they were actually connected to a WWTP. Furthermore, it should be noted that the database does not contain any WWTPs in the vicinity of the Italian city of Napoli. Finally, a set of 26,607 unique combinations of agglomeration and WWTP was constructed.

Due to these missing data on agglomeration connectivity to WWTPs, ePiE predictions are less reliable for Southern Italy and Croatia.

The current ePiE model is set up in such a way that the curated UWWTD database could in principle be updated with future versions of the UWWTD database. This would, however, also require additional curation of the updated database as described in this chapter.

2.2. Fate during wastewater treatment

After APIs are consumed and excreted into wastewater, part of this wastewater might be transported towards a WWTP. In environmental risk assessment of pharmaceuticals, the fraction of an API removed during passage through a WWTP is generally either derived directly from literature (e.g. from measurements of influents and effluents in full scale WWTPs), or estimated using predictive models that simulate the chemical fate in WWTPs. The former approach is not a viable option for the ePiE model developed, because it does not allow for an assessment tailored to the specific characteristics of the local WWTP, nor does it allow predictions for APIs under development. Therefore, the multimedia model SimpleTreat 4.0 (Struijs, 2014; Struijs et al., 2016) was implemented in the ePiE model. This model estimates the partitioning and degradation of a substance in an activated sludge WWTP (i.e. secondary treatment). The model output consists of concentrations in each compartment and phase (air, water and different sludge types) from which the release of pollutants into the environment is derived. It can also be used for the estimation of removal during primary treatment. In that case, only the removal after passage through the primary settling tank is assessed to derive a removal value. While originally spreadsheet-based, the model was rewritten as a function in R in order to keep the system consistent and practical, and enable the use of a single platform for the ePiE model. The ePiE model assumes a default WWTP design as formulated in SimpleTreat 4.0 (Struijs, 2014), except for the size of the WWTP which is derived from the UWWTD database. Local environmental characteristics relevant for the estimation of removal efficiency are surface air temperature and wind speed. These are incorporated at the level of individual WWTPs using spatial information from the Climatic Research Unit (CRU) Time-Series (TS) v3.10 on air temperature (Jones and Harris, 2013) and spatial information from long term yearly mean wind speed at surface (1981-2010) from NCEP/NCAR (Kalnay et al., 1996) on wind speed.

SimpleTreat 4.0 requires the input of three partitioning coefficients, namely the dimensionless air/water partitioning coefficient K_{AW} , the sewage solids-water partitioning coefficient K_{pss}

(L/kg), and the activated sludge solids-water partitioning coefficient Kp_{as} (L/kg). While SimpleTreat 4.0 provides the option to estimate these based on pK_a and a classification of the chemical as neutral, weak acid, or weak base, we decided to separate the parameterisation of partitioning coefficients from the rest of the SimpleTreat 4.0 model. Firstly, this enables consistency with the rest of the modelling framework. Secondly, it ensures that newly developed QSAR/PR/BR models for estimation of partitioning coefficients can be implemented in the iPiE system separately from the ePiE model, keeping their integration flexible and straightforward. To this end, the ePiE model requires all chemical-specific input parameters to be provided both for the neutral form and the alternate (ionized) form of the API. During model calculations these are then combined at local pH. Chapter 4.1 describes this in more detail. After parameterisation, partitioning coefficients are fed into the SimpleTreat 4.0 model. Moreover, SimpleTreat 4.0 also requires the input of a first-order biodegradation rate constant ($k_{bio, WWTP}$; s^{-1}) for the calculation of removal during secondary (activated sludge) treatment. Finally, SimpleTreat 4.0 requires basic physicochemical properties as input, being the molecular weight (MW ; g/mol) (for APIs that are salts, the molecular weight of the free acid/base is required), the water solubility (S_w ; mg/L), and the vapour pressure (P_v ; Pa).

As mentioned in Chapter 2.1, the UWWTD database not only distinguishes between primary and secondary treatment, but also indicates which WWTPs in Europe apply more advanced (tertiary) treatment techniques. Due to insufficient data availability on degradation rates and reactor kinetics for these techniques, the ePiE model currently cannot estimate removal of chemicals via chlorination, UV-treatment, and ozonation. Application of such disinfection techniques in wastewater treatment is currently relatively uncommon in Europe, although it will likely increase in the future. According to the UWWTD database, 13% of European WWTPs applies some form of chlorination, 3% applies some form of UV-treatment, and only 0.2% some sort of ozonation. However, these disinfection techniques seem to achieve quite substantial removal efficiency of some APIs compared to traditional primary and secondary treatment (e.g. Esplugas et al., 2007; Nakada et al., 2007). Consequently, their local presence or absence can substantially contribute to spatial variability in emissions to surface waters. Therefore, emission to surface waters could be overestimated by the ePiE model if the surface water is connected to a WWTP with tertiary treatment techniques.

2.3. Allocation of emission locations

Each WWTP and each agglomeration with less than 100% WWTP connectivity was assigned an emission point, i.e. the specific location where the emission to surface water occurs. These locations form the spatial starting point of the concentration calculations as described in Chapter 3.

River networks in ePiE were constructed for individual drainage basins as delineated by the global database HydroSHEDS (Lehner et al., 2008b). Per basin, its borders were used to crop the river network, available as spatial line objects at 30 arcseconds (~1 km) from the HydroSHEDS database, which were then translated into a binary 30 arcseconds raster with information on the presence or absence of a river element. Next, the raster cells containing river elements were used to create a spatial point file, with each point representing a network node. All nodes were automatically classified as either:

1. junction (node where two streams meet, has two upstream nodes);

2. mouth (node where river flows into the sea);
3. start (nodes representing a river source);
4. regular node.

Via overlay of the river network with the lakes and reservoirs in the global database HydroLAKES (Messenger et al., 2016), available as spatial polygons, nodes located within a lake or reservoir were identified. These nodes are skipped during model computations. Additionally, intersections between the river network and lakes and reservoirs were determined, and were added as additional nodes to the network, classified as either:

1. outlet (the one intersection per lake/reservoir closest to the river mouth);
2. inlet (all other intersections).

Properties associated to the lakes and reservoirs in HydroLAKES, i.e., depth and hydraulic retention time, were added to their corresponding outlet node.

After the curation steps described in Chapter 2.1, the WWTPs and agglomerations from the UWWTD-Waterbase database (EEA, 2015) were snapped to the river network and incorporated as emission sources. Any WWTPs and agglomerations located within lakes or reservoirs were allocated to these as direct emission source. Direct emissions into the sea were excluded from the model. The up- and downstream node for all WWTP and agglomeration specific nodes was determined using its relative placement in the river network.

Using the flow direction raster available from HydroSHEDS, the hydrological interconnectivity of the nodes in the network was determined, including distances between them. Finally, gridded information on air temperature (Jones and Harris, 2013), wind speed (Kalnay et al., 1996), slope (Verdin et al., 2007), and streamflow (Barbarossa et al., 2018) was extracted to all nodes in the network. These calculations were done using R 3.3.1 (R Core Team, 2015).

3. Estimation of concentrations

After consumption and excretion as parent compound to wastewater, APIs enter surface waters either directly, or after incomplete removal during wastewater treatment. The ePiE model then follows the APIs through the network of each river basin. As explained in Chapter 2.3, this river network can be summarized as a collection of different types of nodes and the spatial relationships between them, i.e. their interconnectivity.

The ePiE model follows the APIs through the river network via directed calculations, ordered from the most upstream to the most downstream points (Chapter 3.1). Along the way, the ePiE model accounts for dilution in the water column and five separate dissipation processes m : biodegradation, photolysis, hydrolysis, volatilisation, and sedimentation. These dissipation processes are implemented as (pseudo-) first order rate constants, corrected for the non-adsorbed available fraction. These are described in more detail in Chapter 5.1. The process of sedimentation is also required to estimate concentrations in the sediment layer below the water column (Chapter 3.2).

3.1. API concentrations in rivers and lakes

The ePiE model calculates total concentrations (combined fractions dissolved and bound to suspended solids) of APIs in the water column at all nodes in the network. *Equation 3* calculates concentration C_i ($\mu\text{g/L}$) at any node i in the river network. In addition to concentrations in river water, the ePiE model also calculates concentrations in lake waters. *Equation 4* calculates concentration C_i ($\mu\text{g/L}$) in lake or reservoir i , following an approach similar to Grill et al.¹⁵ in which they are modelled as single completely stirred tank reactors.

$$C_i = \frac{E_{w,i} + \sum_{j=1}^n \left(E_{w,j} \cdot e^{-\left[\sum_{m=1}^5 k_{m,d_{j-i}} \right] \frac{d_{j-i}}{v_{d_{j-i}}}} \right)}{Q_i} \quad \text{Equation 3}$$

- $E_{w,i}$ and $E_{w,j}$ are the emissions into the river network at node i and at node j upstream from node i , respectively (mg/s);
- n is the total number of nodes upstream from node i (-);
- d_{j-i} is the distance over the river network between node j and node i (m);
- $k_{m,d_{j-i}}$ is the average (pseudo-) first order rate constant for loss process m over d_{j-i} (s^{-1}) (Chapter 5);
- $v_{d_{j-i}}$ is the average river flow velocity over d_{j-i} (m/s) (Chapter 3.3);
- Q_i is the river flow at node i (m^3/s) (Chapter 3.3).

$$C_i = \frac{\sum_{p=1}^n (E_{w,p})}{(V_i / HRT_i) + \sum_{m=1}^5 (k_{m,i}) \cdot V_i} \quad \text{Equation 4}$$

- $E_{w,p}$ is the emission into lake or reservoir i coming from node p (mg/s), which can either be a direct emission source (i.e., a WWTP or an agglomeration), or an inlet point carrying API residues from upstream the river network;
- n is the total number of nodes emitting into lake or reservoir i (-);
- HRT_i is the hydraulic retention time of lake or reservoir i (s) (Chapter 3.3);

- V_i is the volume in lake or reservoir i (m^3) (Chapter 3.3);
- $k_{m,i}$ is the (pseudo-) first order rate constant for loss process m in lake or reservoir i (s^{-1}) (Chapter 5).

3.2. API concentrations in sediment

The process of sedimentation results in possible accumulation of pharmaceuticals in the sediment layer below the water column. Total and dissolved (non-sorbed) concentrations in the upper sediment layer of river nodes are calculated via Equation 5 and Equation 6. Total and dissolved concentrations in the upper sediment layer of lakes or reservoirs are also calculated using Equation 5 and Equation 6. Calculation of steady state total concentration in sediments is based on the concentration in the overhead water column (C_i ; Equation 3 and Equation 4). **Prediction of sediment concentrations is currently only included in the R version and not in the iPiEsys version of ePiE because the accuracy of the predicted sediment concentrations has not yet been checked based on a comparison with measurement data.**

The ratio between input of API to the sediment via sedimentation (via k_{sed} ; s^{-1}) and removal from the sediment via biodegradation and hydrolysis (via $k_{bio,sed}$ and $k_{hydro,sed}$; s^{-1}) is accounted for. Then, the volumetric ratio between the water column and the sediment layer is accounted for and determined from their depths H_w and H_{sed} (m). Finally, the porosity of the sediment θ_{sed} (L_{water}/L_{sed}) and its mineral density ρ_{sed} (kg_{solids}/L_{solids}) are used to transform the concentration per volume sediment to a concentration per kilogram sediment. The mineral density $\rho_{s,sed}$ has to be corrected for the volume fraction of the solids (i.e. $1 - \theta_{sed}$) because it relates to the mass per volume of solids and not the total sediment. The fraction of API present as dissolved chemical in the upper sediment layer ($f_{sed,diss}$; -) is used to estimate the steady state dissolved concentration in the upper sediment layer $C_{sed,tot}$ ($\mu g/kg$) (Chapter 5).

$$C_{sed,tot} = C_{rw,tot} \cdot \frac{k_{sed}}{k_{bio,sed} + k_{hydro,sed}} \cdot \frac{H_w}{H_{sed}} \cdot (\theta_{sed} \cdot \rho_w + (1 - \theta_{sed}) \cdot \rho_{s,sed}) \quad \text{Equation 5}$$

$$C_{sed,diss} = f_{sed,diss} \cdot C_{sed,tot} \quad \text{Equation 6}$$

- $C_{sed,tot}$ is the steady state total concentration in the upper sediment layer ($\mu g/kg$);
- $C_{sed,diss}$ is the steady state dissolved concentration in the upper sediment layer ($\mu g/kg$);
- $C_{rw,tot}$ is the total river water concentration resulting from Equation 3 or Equation 4 ($\mu g/L$);
- k_{sed} is the first order sedimentation rate constant (s^{-1}) (Chapter 5.1);
- $k_{bio,sed}$ is the pseudo-first order biodegradation rate constant in sediment (s^{-1}) (Chapter 5.1);
- $k_{hydro,sed}$ is the first order hydrolysis rate constant in sediment (s^{-1}) (Chapter 5.1);
- H_w is the depth of the water column (m) (Chapter 3.3);
- H_{sed} is the thickness of the upper sediment layer, with a fixed default value of 0.03 (m);
- θ_{sed} is the porosity of sediment, with a fixed default of 0.8 (Paterson and Mackay, 1995) (L_{water}/L_{sed});
- ρ_w is the density of water, with a fixed default of 1 (kg_{water}/L_{water});
- $\rho_{s,sed}$ is the mineral density of sediment solids, with a fixed default of 2.1633 (Fantke et al., 2016; Honti et al., 2016) (kg_{solids}/L_{solids});
- $f_{sed,diss}$ is the fraction of API present as dissolved chemical in the upper sediment layer (-) (Chapter 5.1).

In Equation 3 and Equation 4, the total number of nodes (n ; -) and the distance over the river network between node j and node i (d_{j-i} ; m) are determined using HydroSHEDS. Hydrological parameters such as discharge volumes, flow velocities, hydraulic retention times, and dimensions of rivers and lakes, are required to calculate concentrations in surface waters and sediments. The hydrological parameterisation of the river network is described in Chapter 3.3. Calculations of degradation rate constants in Equation 3, Equation 4, Equation 5 and Equation 6 are described in Chapter 5.

3.3. Hydrology of rivers and lakes

For a location-specific estimation of API concentrations, the ePiE model incorporates a methodology to hydrologically characterise all nodes throughout the European river network. More specifically, for river nodes the following hydrological characteristics are required: river discharge volumes (Q ; m³/s), river flow velocities (v ; m/s), and river depth (H_w ; m) (as applied in Equation 3 and Equation 5). For lakes and reservoirs, hydraulic retention times (HRT ; s), and lake depth (H ; m) are required (as applied in Equation 4 and Equation 5).

River discharge volume Q

In the ePiE model, year-specific flow metrics are automatically extracted to the nodes in the river network, depending on the year and flow conditions selected by the user. For characterization of annual mean flow, and highest and lowest monthly mean flow, the global FLO1K dataset (Barbarossa et al., 2018) was implemented in ePiE. FLO1K is based on an ensemble of artificial neural networks regressions, with upstream-catchment physiography (area, slope, elevation) and year-specific climatic variables (precipitation, temperature, potential evapotranspiration, aridity index and seasonality indices) as covariates. It provides estimations of flow at a spatial resolution of 30 arc seconds (~1 km) for the years 1960-2015, which are in good agreement with independent data (global R^2 of single-year metrics up to 0.91). An additional comparison with independent data obtained from 1,007 European monitoring stations for the period 2010-2015 (GRDC, 2015), showed that year-specific annual mean flow, and highest and lowest mean monthly flow in European rivers are predicted well, with R^2 values of 0.97, 0.95 and 0.91, respectively (Figure 2).

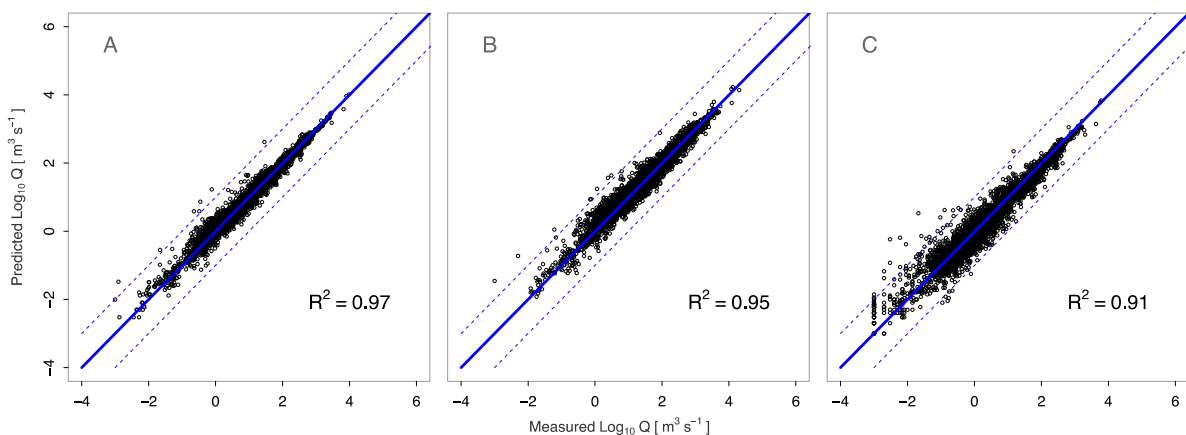


Figure 2. Validation results for year-specific annual mean (A), maximum (B) and minimum (C) monthly flow. Independent validation dataset consisted of yearly measurements (2010-2015) from 1,007 GRDC European stations. The solid line represents perfect model fit (1:1 line) and the dashed lines represent a difference of one order of magnitude.

River flow velocity v and river depth H_w

River flow velocity is commonly estimated with the widely used Manning-Strickler equation for open channel flow (Equation 7) (e.g. Pistocchi and Pennington, 2006; Schulze et al., 2005).

$$v = \frac{R^{2/3} \cdot S^{1/2}}{n} \quad \text{Equation 7}$$

- v is the river's flow velocity (m/s);
- R is the hydraulic radius of the river (m);
- S is the river slope (m/m);
- n is the river bed roughness (s/m^{1/3}).

The Manning-Strickler equation makes use of the hydraulic radius R , which is defined as the ratio of the cross-sectional area A (m²) over the wetted perimeter P (m): $R=A/P$. For estimation of this hydraulic radius R , it is often assumed that the river bed is shaped as a rectangle. Moreover, since the depth of the river's water column H_w (m) is generally much smaller than the width of the river (W ; m), especially for major rivers, R can be approximated by depth (Pistocchi and Pennington, 2006):

$$R = \frac{A}{P} = \frac{H_w \cdot W}{2 \cdot H_w + W} \cong \frac{H_w \cdot W}{W} = H_w \quad \text{Equation 8}$$

- A is the cross-sectional area of the river (m²);
- P is the wetted perimeter of the river (m);
- H_w is the depth of the river's water column (m);
- W is the width of the river (m).

Additionally, the assumption of a rectangular river bed means that the flow velocity v is related to the discharge volume Q , river depth H_w , and river width W (Equation 9), enabling the calculation of river depth via Equation 10.

$$v = \frac{Q}{A} = \frac{Q}{H_w \cdot W} \quad \text{Equation 9}$$

$$H_w = \frac{Q}{v \cdot W} \quad \text{Equation 10}$$

Thus, R in the Manning-Strickler equation (Equation 7) can be substituted by the term $\frac{Q}{v \cdot W}$, resulting in Equation 11.

$$v = n^{-3/5} \cdot Q^{2/5} \cdot W^{-2/5} \cdot S^{3/10} \quad \text{Equation 11}$$

Consequently, in order to estimate flow velocity v , four parameters are minimally required: river discharge volume Q , river width W , river bed roughness n , and river slope S (m/m). For Q , we used location specific estimations via the spatial regression model described above, and for S we used the HydroSHEDS Digital Elevation Map (DEM) at 3 arc-seconds resolution (Lehner et al., 2008a).

For the other two parameters, i.e. river width W and river bed roughness n , continuous spatially explicit data are generally not available. Andreadis et al. (2013) calculated width and

depth for all streams in the HydroSHEDS database. However, they used relatively poor estimations of discharge, and one overall global equation describing the relationships between discharge and width and between discharge and depth, without any spatial differentiation. Since the relationship between discharge and river dimensions varies spatially, we applied Equation 12 for river width of European rivers specifically, as derived by Pistocchi and Pennington (2006) ($r^2=0.868$). They based this regression on 186 river width measures across the European continent covering a range of ~10 m to ~1000 m.

$$W = 7.3607 \cdot Q^{0.52425} \quad \text{Equation 12}$$

In addition to a European average equation, Pistocchi and Pennington (2006) also derived regional equations for specific river catchments in Europe. However, the spatial extent of these regressions (i.e. where does which regression apply), does not become specifically clear from the reported information. They are merely indicatively described (e.g. “East of the Rhine”), and therefore not usable for the ePiE model.

Values for Manning’s roughness coefficient n in natural streams generally vary between 0.015 s/m^{1/3} and 0.07 s/m^{1/3} for flows with less than bankfull discharge. They can reach up to 0.25 for overbank flows (Schulze et al., 2005). Very few local data exist, and assessing local river bed roughness throughout Europe is thus impossible. Based on Pistocchi and Pennington (2006), we applied an overall value for n of 0.045 s/m^{1/3}. This value represents optimality: using it ensures that errors on the estimated velocity are maximally limited, and approximately within ±40% for any “true” value of n that might occur locally.

4. Partitioning coefficients

The partitioning of APIs between a range of media and water is incorporated at different points throughout the ePiE model. Partitioning coefficients are estimated via the weighted summation of partitioning coefficients for the separate forms of the API (neutral and alternate, i.e. anionic or cationic). This is done in order to allow for location-specific parameterisation due to geographical differences in aquatic pH. However, the potential for operationalization of a spatially explicit approach is currently still limited due to lacking spatially explicit data on environmental properties such as carbon content, temperature and pH.

The weighted summation is based on the Henderson-Hasselbalch equation (Equation 13), which calculates the fraction of the chemical present in its neutral form from the pH of the water assessed and the dissociating properties of the compound. Logically, Equation 13 does not apply to non-ionizing neutral compounds, for which the neutral fraction equals 1 regardless of the pH of the medium assessed.

$$f_n = \frac{1}{(1 + 10^{a \cdot (pH - pK_a)})} \quad \text{Equation 13}$$

- f_n is the fraction of the chemical present in its neutral form (-);
- a is a factor that has value 1 for anionic compounds, and -1 for cationic compounds;
- pH is the acidity of the water assessed (-; default 7.4 for sediments and surface water and default 7.0 for primary and secondary treatment);
- pK_a is the acid dissociation constant of the chemical (-) (user input).

Subsequently, the neutral fraction is multiplied with the partitioning coefficient for the neutral form of the API, and the ionized fraction (i.e. $1 - f_n$) with the partitioning coefficient of the alternate, dissociated form of the API. These are then summed into one value for the total compound's partitioning coefficient.

4.1. Solids-water partitioning

The ePiE model requires the input of solids-water partitioning (Kp ; L/kg) values for five different types of solids-water matrices: primary sewage, secondary sludge, suspended matter, dissolved organic carbon, and sediments. Separate partitioning coefficients are required for the neutral and alternate form of the compound of interest. When no direct Kp values are provided as input (either for the total compound or for the separate neutral/ionized forms), they are currently estimated under the assumption that sorption is solely driven by organic carbon content of the medium assessed. Consequently, sorption of the total compound Kp_{tot} is compiled from the organic-carbon water partitioning coefficients of the individual forms ($K_{OC,n}$, $K_{OC,alt}$; kg/L; see QSARs below), the volume fraction organic carbon in the solids (f_{OC} ; -), and f_n :

$$Kp_{tot} = (f_n \cdot K_{OC,n} + (1 - f_n) \cdot K_{OC,alt}) \cdot f_{OC} \quad \text{Equation 14}$$

- Kp_{tot} is the partitioning coefficient between solids and water for the total compound (L_{water}/kg_{solids});
- f_n is the fraction of the chemical present in its neutral form (-) (Equation 13);
- $K_{OC,n}$ is the organic-carbon water partitioning coefficient of the neutral form (L/kg);

- $K_{OC,alt}$ is the organic-carbon water partitioning coefficient of the alternate (anionic or cationic) form (L/kg);
- f_{OC} is the volume fraction of organic carbon in the solids assessed (-; defaults: 0.3 for primary sludge, 0.37 for secondary sludge and 0.05 for sediment).

K_{OC} values of the individual forms can be provided directly, or be derived via the application of QSAR models from literature. For neutral compounds, the QSAR derived by Sabljčić et al. (1995) (Equation 15) is applied, and for anionizing compounds Equation 16 ($K_{OC,n}$) and Equation 17 ($K_{OC,alt}$) are applied (Franco and Trapp, 2008). QSARs from Franco and Trapp (2008) are also applied for cationic compounds: Equation 18 for their $K_{OC,n}$ and Equation 19 for their $K_{OC,alt}$.

$$K_{OC,n} = 1.26 \cdot K_{OW,n}^{0.81} \quad \text{Equation 15}$$

$$K_{OC,n} = 10^{0.54 \cdot \log(K_{OW,n}) + 1.11} \quad \text{Equation 16}$$

$$K_{OC,alt} = 10^{0.11 \cdot \log(K_{OW,n}) + 1.54} \quad \text{Equation 17}$$

$$K_{OC,n} = 10^{0.37 \cdot \log(K_{OW,n}) + 1.70} \quad \text{Equation 18}$$

$$K_{OC,alt} = 10^{pK_a^{0.65} \left(\frac{K_{OW,n}}{K_{OW,n} + 1} \right)^{0.14}} \quad \text{Equation 19}$$

- $K_{OC,n}$ is the organic-carbon water partitioning coefficient for the neutral form (L_{water}/kg_{OC});
- $K_{OC,alt}$ is the organic-carbon water partitioning coefficient for the alternate (anionic or cationic) form (L_{water}/kg_{OC});
- $K_{OW,n}$ is the octanol water partitioning coefficient for the neutral form (-) (user input);
- pK_a is the acid dissociation constant of the chemical (-) (user input).

It is generally agreed upon that the performance of currently available models for the estimation of solids-water partitioning coefficients is relatively poor, especially those for ionizing compounds. Table 1 provides an overview of the performance and application domain of the different QSARs implemented in ePiE.

Table 1. Partitioning coefficients used in ePiE that are based on QSARs, and model restrictions related to these QSARs.

Parameter estimated by QSAR	QSAR(s) in Technical model description (v1)	QSAR application domain	Limitations in application domain	Reliability	Reference
$K_{OC,n}$	15	QSAR for neutral compounds. Based on data from 81 hydrophobic chemicals with $\log K_{ow}$ from 1 to 7.5. The equation had an r^2 of 0.887.	During validation, the smallest errors occurred for a $\log K_{ow}$ of 1 to 4.	The equation is considered reasonably reliable.	(Sabljčić et al., 1995)
$K_{OC,n}$ and $K_{OC,alt}$	16 and 17	QSAR for anionizing compounds. Based on data from 93 acids with a pK_a between 0 and 12 and a $\log K_{ow,n}$ of -2 to 8. Equation 16 had an r^2 of 0.81 (n=11) and Equation 17 an r^2 of 0.04 (n=29). The combined equation had an r^2 of 0.54 for the training set (n=62) and an r^2 of 0.44 for the validation set (n=31)	Can introduce considerable uncertainty in estimating partitioning behaviour of acids, particularly strong acids.	QSAR performance is suboptimal. If available, empirical K_{oc} values measured at the relevant pH should be preferred.	(Franco and Trapp, 2008)
$K_{OC,n}$ and $K_{OC,alt}$	18 and 19	QSAR for cationic compounds. Based on data from 65 bases with a pK_a	Can introduce considerable uncertainty in estimating	QSAR performance is	(Franco and

		between 2 and 12 and a logK _{ow,n} of -2 to 7. Equation 38 had an r ² of 0.83 (n=11) and Equation 37 an r ² of 0.75 (n=13). The combined equation had an r ² of 0.60 for the training set (n=43) and an r ² of 0.44 for the validation set (n=22).	partitioning behaviour of bases.	suboptimal. If available, empirical K _{oc} values measured at the relevant pH should be preferred.	Trapp, 2008)
<i>K_{pDOC}</i>	20	QSAR for neutral compounds. Based on 127 data points from 14 nonionizable chemicals.	Can introduce considerable uncertainty in estimating partitioning behaviour.	If available, empirical K _{pDOC} values should be preferred.	(Burkhard, 2000)
<i>K_{OW,app}</i>	21	QSAR for ionizing compounds. The equation was based on a very limited data set of carboxylic acid and 4 other anions.	Equation is based on the assumption that the K _{ow} of the ionized form is approximately a factor of 3.5 lower than that of the neutral form.	Empirical basis is very weak and empirical data should always be preferred over this QSAR estimate.	(Trapp and Horobin, 2005)

It is anticipated that in due time, more sophisticated models for *K_p* estimation will become available. This will result in substitution of the currently implemented QSARs, and might require expansion of the current approach to include other constituents of the matrix assessed in addition to organic carbon. For example, the clay content of the medium and its cation exchange capacity has been shown relevant for sorption of cations (Droge and Goss, 2013).

For sorption to dissolved organic carbon in the water column, characterised with *K_{pDOC}* (L/kg), a different approach is followed. When no direct input value is provided, it is estimated directly from the apparent octanol water partitioning coefficient (*K_{OW,app}*), using Equation 20 as derived by Burkhard (2000). For ionizing chemicals, *K_{OW,app}* is calculated with Equation 21, based on the apparent fractions of the neutral and alternate forms of the chemical (Trapp and Horobin, 2005).

$$K_{pDOC} = 0.08 \cdot K_{OW,app} \quad \text{Equation 20}$$

$$K_{OW,app} = f_n \cdot K_{OW,n} + (1 - f_n) \cdot 10^{\log(K_{OW,n}) - 3.5} \quad \text{Equation 21}$$

- *K_{pDOC}* is the dissolved organic carbon-water partitioning coefficient (L/kg);
- *K_{OW,app}* is the apparent octanol water partitioning coefficient (-).

4.2. Air-water partitioning

Contrary to the approach followed for solids-water partitioning, we did not distinguish between individual chemical forms in the estimation of air-water partitioning coefficients (*K_{AW}*; -). Instead, the total *K_{AW}* is estimated regardless of ionization of the compound. For APIs we expect this simplification to be of little influence, considering the relatively small role that volatilisation has for these generally non-volatile compounds.

$$K_{AW} = \frac{P_v \cdot MW}{S_w \cdot R \cdot T} \quad \text{Equation 22}$$

- *K_{AW}* is the dimensionless air-water partitioning coefficient (-);
- *P_v* is the vapour pressure of the API under consideration at 25 °C (Pa) (user input; default value 1.00E-10 Pa);
- *MW* is the molecular weight of the API under consideration (g/mol) (for APIs that are salts, the molecular weight of the free acid/base is required) (user input);

- S_w is the solubility in water of the API under consideration at 25 °C (mg/L) (user input; default value 1.00E+03 mg/L);
- R is the universal gas constant (Pa/m³/mol/K), with a standard value of 8.314 Pa/m³/mol/K;
- T is the local water temperature (K) (user input; default value 285K).

5. Environmental dissipation processes

The ePiE model accounts for five separate processes that result in dissipation of APIs from environmental media, being the water column and the upper sediment layer. Three of these five dissipation processes are degradation processes, i.e. biodegradation, photolysis and hydrolysis; two of the five dissipation processes are advective (transport) processes, i.e. sedimentation (water to sediment) and volatilisation (water to air). The degradation processes are described in Chapter 5.1, the advective processes in Chapter 5.2.

5.1. Degradation processes

Within the ePiE model, biodegradation, photolysis and hydrolysis are all assumed to occur in surface waters (Equation 3 and Equation 4). In sediments, photolysis is assumed negligible, and only biodegradation and hydrolysis are considered (Equation 5). The model requires separate input of degradation rate constants for the ionized and neutral form of the API. In practice, separate values for the ionized and alternate forms are rarely available and the same value is being used for both. If without any input, degradation is assumed to be zero. The degradation rate constants of the individual processes are aggregated into a total degradation rate constant. Degradation rate constants are implemented as follows:

- Second order biodegradation rate constant k_{bio} incorporates the concentration of bacterial cells in the water and is expressed as $L_{water}/cells/s$. In practice, information on bacterial cell density is often lacking from tests as well as the environment. Experimental (pseudo-)first order degradation rates in surface water are then directly used under the assumption of equal bacterial cell densities between test and field;
- First order photolysis rate constant $k_{photo,total}$ is expressed as s^{-1} . The ePiE model allows the input of an overall photolysis rate constant (combined direct and indirect photolysis);
- First order hydrolysis rate constant k_{hydro} is expressed as s^{-1} .

Moreover, we conservatively assumed that degradation processes only relate to the available fraction of the API and that the fraction adsorbed to solids is not available for degradation. Therefore, degradation rate constants for the individual processes are corrected for sorption via fraction dissolved f_{diss} (-) (e.g. Honti et al., 2016). For degradation processes occurring in surface waters, this is done via Equation 23 and for sediments via Equation 24:

$$f_{w,diss} = \frac{1}{1 + Kp_{tot,susp} \cdot C_{susp} + Kp_{tot,DOC} \cdot C_{DOC}} \quad \text{Equation 23}$$

$$f_{sed,diss} = \frac{1}{1 + Kp_{tot,sed} \cdot \rho_{s,sed} \cdot \frac{(1 - \theta_{sed})}{\theta_{sed}}} \quad \text{Equation 24}$$

- $f_{w,diss}$ is the fraction of API present as dissolved chemical in the water column (-);
- $f_{sed,diss}$ is the fraction of API present as dissolved chemical in the upper sediment layer (-);
- $f_{soil,diss}$ is the fraction of API present as dissolved chemical in agricultural soil (-);
- $Kp_{tot,susp}$ is the partitioning coefficient between water and suspended solids for the total compound (L_{water}/kg_{solids}) (optional user input or Equation 14-19);
- $Kp_{tot,DOC}$ is the partitioning coefficient between water and DOC for the total compound (L_{water}/kg_{DOC}) (optional user input or Equation 20-21);
- $Kp_{tot,sed}$ is the partitioning coefficient between water and sediment solids for the total compound (L_{water}/kg_{solids}) (optional user input or Equation 14-19);

- C_{susp} is the concentration suspended solids in the water column, with a fixed default of $0.015 \cdot 10^{-3}$ (Humbert et al., 2011) ($\text{kg}_{\text{solids}}/\text{L}_{\text{water}}$);
- C_{DOC} is the concentration dissolved organic carbon in the water column, with a fixed default of $0.005 \cdot 10^{-3}$ (Asselman, 1997) ($\text{kg}_{\text{DOC}}/\text{L}_{\text{water}}$);
- $\rho_{s, sed}$ is the mineral density of sediment solids, with a fixed default of 2.1633 (Fantke et al., 2016; Honti et al., 2016) ($\text{kg}_{\text{solids}}/\text{L}_{\text{solids}}$);
- θ_{sed} is the porosity of sediment, with a fixed default of 0.8 (Paterson and Mackay, 1995) ($\text{L}_{\text{water}}/\text{L}_{\text{sed}}$);

In addition to a correction for sorption, individual degradation processes are also corrected for differences between local temperature and temperature during degradation testing. Using the Arrhenius equation to account for the temperature dependence of the degradation rate constants, a local correction factor f_{temp} is derived (Equation 25).

$$f_{temp} = 2^{((T_{field} - T_{test})/10)} \quad \text{Equation 25}$$

- f_{temp} is the local correction factor for temperature (-);
- T_{test} is the temperature at which the degradation rate constant was derived (K) (user input; default value 293.15 K);
- T_{field} is the local temperature at which the degradation rate constant is applied, with a default of 285 (ECHA, 2016) when georeferenced water temperatures are not available (K).

In addition to a correction for temperature, photolysis rate constants are corrected for reduced light intensity with water depth. This correction for water depth is done with correction factor f_{depth} (-), based on Schwarzenbach et al. (1993) (Equation 26).

$$f_{depth} = \frac{1 - 10^{-D_{\lambda} \cdot \alpha_{\lambda} \cdot 100 \cdot H_w}}{\ln(10) \cdot D_{\lambda} \cdot \alpha_{\lambda} \cdot 100 \cdot H_w} \quad \text{Equation 26}$$

- f_{depth} is the local correction factor for water depth (-);
- D_{λ} is the ratio between the average path length of light with wavelength λ , and the average water depth (-), with a proposed default value for non-turbid waters of 1.2 (Schwarzenbach et al., 1993);
- α_{λ} is the beam attenuation coefficient for light at wavelength λ (cm^{-1}) (Table 2);
- H_w is the average depth of the water column (m) (Equation 10).

Table , copied from Table 13.6 in Schwarzenbach et al. (1993), contains values for α_{λ} at a range of wavelengths relevant for solar radiation. This tabular relationship is applicable to a clear midsummer day at 47.5 °N, and was used to assign values for α_{λ} to a chemical based on its maximum absorption wavelength λ_{max} (nm). A default value of 298 nm for the intensity of solar radiation can be used. At 298 nm the attenuation coefficient is highest, i.e. 0.0430 cm^{-1} . The highest attenuation coefficient results in the lowest light penetration in water, hence a conservative estimate of photolysis.

Table 2. Average beam attenuation coefficients and intensity of solar radiation at wavelengths relevant for solar radiation.

λ (nm) (min-max)	Average α_{λ} (cm^{-1})
296.25 - 298.75	0.0430

298.75 - 301.25	0.0415
301.25 - 303.75	0.0395
303.75 - 306.25	0.0375
306.25 - 308.75	0.0355
308.75 - 311.25	0.0335
311.25 - 313.75	0.0320
313.75 - 316.25	0.0305
316.25 - 318.75	0.0290
318.75 - 321.25	0.0275
321.25 - 325	0.0260
325 - 335	0.0220
335 - 345	0.0185
345 - 355	0.0150
355 - 365	0.0125
365 - 375	0.0100
375 - 385	0.0083
385 - 395	0.0069
395 - 405	0.0055
405 - 435	0.0042
435 - 465	0.0028
465 - 495	0.0019
495 - 600	0.0010

The locally adapted (pseudo-)first order rate constants for biodegradation ($k_{bio,i}$; s^{-1}), photolysis ($k_{photo,i}$; s^{-1}), and hydrolysis ($k_{hydro,i}$; s^{-1}) are estimated via Equation 27, Equation 28, and Equation 29, respectively.

$$k_{bio,i} = k_{bio} \cdot f_{temp} \cdot f_{diss} \quad \text{Equation 27}$$

$$k_{photo,i} = k_{photo,total} \cdot f_{light} \cdot f_{temp} \cdot f_{depth} \cdot f_{diss} \quad \text{Equation 28}$$

$$k_{hydro,i} = k_{hydro} \cdot f_{temp} \cdot f_{diss} \quad \text{Equation 29}$$

- $k_{bio,i}$ is the pseudo-first order biodegradation rate constant adapted to conditions at i (s^{-1});
- $k_{photo,i}$ is the pseudo-first order photolysis rate constant adapted to conditions at i (s^{-1});
- $k_{hydro,i}$ is the pseudo-first order hydrolysis rate constant adapted to conditions at i (s^{-1});
- k_{bio} is the experimentally determined pseudo-first order biodegradation rate constant (s^{-1}) (user input; default 0);
- $k_{photo,total}$ is the experimentally determined pseudo-first order photolysis rate constant (s^{-1}) (user input; default 0);
- k_{hydro} is the experimentally determined pseudo-first order hydrolysis rate constant (s^{-1}) (user input; default 0);
- f_{diss} is the fraction of API present as dissolved chemical (-);
- f_{temp} is the correction factor for temperature (-);
- f_{depth} is the correction factor for water depth (-);
- f_{light} is the time fraction of light per day (-; default 0.5).

5.2. Advective processes

Two advective dissipation processes are incorporated in the ePiE model, being sedimentation from the water column to the sediment, and volatilisation from the water column to air. These processes are modelled via implementation of transport velocities between media, similar to the USEtox 2.0 model (Fantke et al., 2016), according to Margni et al. (2004). Figure 4 provides a schematic visualisation of the processes implemented. From the figure it must be realized that back transportation from air to water is currently not included in these equations (e.g. via gas absorption and wet and dry deposition). Because of the high resolution of the ePiE model this would involve extensive assumptions on processes such as long distance advective transport, the extent of wet and dry deposition, as well as other properties of the air column at specific locations. Moreover, volatilization to air likely plays a minor role in the fate of APIs, which have a relatively low vapour pressure.

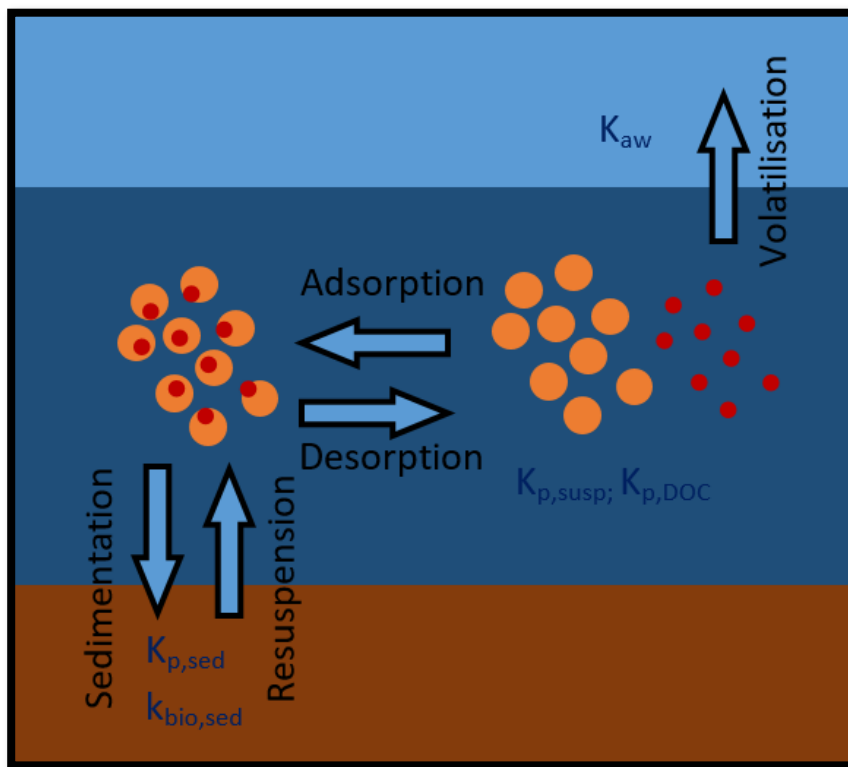


Figure 4. Visual representation of air-water-sediment system and advective transport processes incorporated in the ePiE model.

For volatilisation, the first order volatilisation rate constant from water is calculated from the volatilisation velocity (Equation 30, Equation 31, Equation 32, and Equation 33):

$$k_{vol} = \frac{v_{vol}}{H_w} \quad \text{Equation 30}$$

$$v_{vol} = \frac{v_{m,a,a|w} \cdot v_{m,w,a|w}}{v_{m,a,a|w} \cdot K_{AW} + v_{m,w,a|w}} \cdot K_{AW} \cdot f_{w,diss} \quad \text{Equation 31}$$

- k_{vol} is first order volatilisation rate constant from water to air (s^{-1});
- H_w is the average depth of the water column (m) (Equation 10);
- v_{vol} is the volatilization velocity from water to air (m/s);

- $v_{m,a,a|w}$ is the partial mass transfer coefficient at the air side of the air/water interface (m/s);
- $v_{m,w,a|w}$ is the partial mass transfer coefficient at the water side of the air/water interface (m/s);
- K_{AW} is the dimensionless air-water partitioning coefficient (-) (Equation 22);
- $f_{w,diss}$ is the fraction of API present as dissolved chemical in the water column (-) (Equation 23).

$$v_{m,a,a|w} = 0.01 \cdot (0.3 + 0.2 \cdot v_{wind}) \cdot \frac{18^{(0.67 \cdot 0.5)}}{MW} \quad \text{Equation 32}$$

$$v_{m,w,a|w} = 0.01 \cdot (0.0004 + 0.00004 \cdot v_{wind}^2) \cdot \frac{32^{(0.5 \cdot 0.5)}}{MW} \quad \text{Equation 33}$$

- $v_{m,a,a|w}$ is the partial mass transfer coefficient at the air side of the air/water interface (m/s);
- $v_{m,w,a|w}$ is the partial mass transfer coefficient at the water side of the air/water interface (m/s);
- v_{wind} is the wind speed at the surface (m/s);
- MW is the molecular weight of the API under consideration (g/mol) (for APIs that are salts, the molecular weight of the free acid/base is required) (user input).

Similar to volatilisation, the ePiE model directly accounts for transport of API between water and sediment via adsorption/desorption and sedimentation/resuspension of suspended particulate matter, similar to the USEtox 2.0 model (Fantke et al., 2016), according to Margni et al. (2004). It requires the estimation of adsorption/desorption velocities as well as sedimentation/resuspension velocities, which are combined into one equation describing the rate of sedimentation (Equation 34). The location-specific velocities for the four processes are estimated via Equation 35 (adsorption velocity v_{ads} ; m/s), Equation 36 (desorption velocity v_{des} ; m/s), Equation 37 (sedimentation velocity v_{sed} ; m/s), and Equation 38 (resuspension velocity v_{res} ; m/s). If without any input, degradation rate constants $Kp_{tot,susp}$ and $Kp_{tot,sed}$ will be estimated automatically by ePiE based on K_{oc} .

The sediment phase is treated as a homogeneous phase, consisting of a water sub-phase and a solid sub-phase. Equilibrium is assumed between the pore water and solid sub-phases of the sediment phase. The top layer of the sediment is considered to be well-mixed. If the sedimentation of particles from the water column is greater than the resuspension (net sedimentation), this top layer is continuously being refreshed. The older sediment layer, and with it the chemicals that are associated with the sediment, then gets buried under the freshly deposited material. This is described in Equation 39 and Equation 40.

$$k_{sed} = \left(\frac{v_{ads} + v_{sed}}{H_w} \right) - \frac{\left(\frac{v_{ads} + v_{sed}}{H_w} \right) * \left(\frac{v_{res} + v_{des}}{H_{sed}} \right)}{\left(\frac{v_{res} + v_{des} + v_{sed,acc}}{H_{sed}} + k_{bio,sed,i} \right)} \quad \text{Equation 34}$$

- k_{sed} is the first order sedimentation rate constant from water to sediment (s^{-1});
- v_{ads} is the adsorption velocity from water to sediment (m/s);
- v_{sed} is the sedimentation velocity from water to sediment (m/s);

- v_{des} is the desorption velocity from sediment to water (m/s);
- v_{res} is the resuspension velocity from sediment to water (m/s);
- $v_{sed,acc}$ is the net sediment accumulation rate in water, with a fixed default of $8.6 \cdot 10^{-11}$ (m/s) (Fantke et al., 2016);
- H_w is the depth of the water column (m) (Equation 10);
- H_{sed} is the thickness of the sediment layer, with a default value of 0.03 (m);
- $k_{bio,sed,i}$ is the biodegradation rate in sediment adapted to local conditions (s^{-1}) (In the absence of an experimental value, this rate is estimated from Equation 27 with an extrapolation factor of 0.1 (Rosenbaum et al., 2008)), a correction factor for temperature (Equation 25), and a correction factor for sorption (Equation 24).

$$v_{ads} = \frac{v_{m,w,w|sed} \cdot v_{m,sed,w|sed}}{v_{m,w,w|sed} + v_{m,sed,w|sed}} \cdot f_{w,diss} \quad \text{Equation 35}$$

- v_{ads} is the adsorption velocity from water to sediment (m/s);
- $v_{m,w,w|sed}$ is the partial mass transfer coefficient at the water side of the water/sediment interface, with a fixed default of $2.778 \cdot 10^{-6}$ (Mackay, 2001) (m/s);
- $v_{m,sed,w|sed}$ is the partial mass transfer coefficient at the sediment side of the water/sediment interface, with a fixed default of $2.778 \cdot 10^{-8}$ (Mackay, 2001) (m/s);
- $f_{w,diss}$ is the fraction of the chemical truly dissolved in the surface water (-) (Equation 23).

$$v_{des} = \frac{v_{m,w,w|sed} \cdot v_{m,sd,w|sed}}{v_{m,w,w|sed} + v_{m,sd,w|sed}} \cdot \frac{1}{(\theta_{sed} + (1 - \theta_{sed}) \cdot \rho_{s,sed} \cdot Kp_{tot,sed})} \quad \text{Equation 36}$$

- v_{des} is the desorption velocity from sediment to water (m/s);
- $v_{m,w,w|sed}$ is the partial mass transfer coefficient at the water side of the water/sediment interface, with a fixed default of $2.778 \cdot 10^{-6}$ (Mackay, 2001) (m/s);
- $v_{m,sed,w|sed}$ is the partial mass transfer coefficient at the sediment side of the water/sediment interface, with a fixed default of $2.778 \cdot 10^{-8}$ (Mackay, 2001) (m/s);
- θ_{sed} is the porosity of sediment, with a fixed default of 0.8 (Paterson and Mackay, 1995) (L_{water}/L_{sed});
- $\rho_{s,sed}$ is the mineral density of the sediment, with a fixed default of 2.1633 (Fantke et al., 2016; Honti et al., 2016) (kg_{solids}/L_{solids});
- $Kp_{tot,sed}$ is the sediment-water partitioning coefficient for the total compound (L_{water}/kg_{solids}) (optional user input).

$$v_{sed} = (1 - \theta_{sed}) \cdot v_{sed,gross} \cdot \rho_{s,sed} \cdot Kp_{tot,susp} \cdot f_{w,diss} \quad \text{Equation 37}$$

- v_{sed} is the sedimentation velocity from water to sediment (m/s);
- θ_{sed} is the porosity of sediment, with a fixed default of 0.8 (Paterson and Mackay, 1995) (L_{water}/L_{sed});
- $v_{sed,gross}$ is the gross sedimentation rate from water to sediment (m/s);
- $\rho_{s,sed}$ is the mineral density of the sediment, with a fixed default of 2.1633 (Fantke et al., 2016; Honti et al., 2016) (kg_{solids}/L_{solids});
- $Kp_{tot,susp}$ is the suspended solids-water partitioning coefficient for the total compound (L_{water}/kg_{solids}) (user input);
- $f_{w,diss}$ is the fraction of the chemical truly dissolved in the surface water (-) (Equation 23).

$$v_{res} = v_{sed,gross} - v_{sed,acc} \quad \text{Equation 38}$$

- v_{res} is the resuspension velocity from sediment to water (m/s);
- $v_{sed,gross}$ is the gross sedimentation rate from water to sediment (m/s);
- $v_{sed,acc}$ is the net sediment accumulation rate in water (m/s).

$$\text{If } v_{sed,susp} \cdot \frac{C_{susp}}{\theta_{sed} \cdot \rho_w + (1 - \theta_{sed}) \cdot \rho_{s,sed}} > v_{sed,acc} \quad \text{then:}$$

$$v_{sed,gross} = v_{sed,susp} \cdot \frac{C_{susp}}{\rho_{sd,bulk}}, \text{ else:} \quad \text{Equation 39}$$

$$v_{sed,gross} = v_{sed,acc}$$

- $v_{sed,susp}$ is the settling velocity of suspended particles, with a fixed default $2.89 \cdot 10^{-5}$ (den Hollander et al., 2004) (m/s);
- C_{susp} is the concentration suspended solids in the water column, with a fixed default of $0.015 \cdot 10^{-3}$ (Humbert et al., 2011) ($\text{kg}_{\text{solids}}/\text{L}_{\text{water}}$);
- θ_{sed} is the porosity of the sediment, with a fixed default of 0.8 (Paterson and Mackay, 1995) ($\text{L}_{\text{water}}/\text{L}_{\text{sed}}$);
- ρ_w is the density of water ($\text{kg}_{\text{water}}/\text{L}_{\text{water}}$), with a standard value of 1 kg/L;
- $\rho_{s,sed}$ is the mineral density of the sediment, with a fixed default of 2.1633 (Fantke et al., 2016; Honti et al., 2016) ($\text{kg}_{\text{solids}}/\text{L}_{\text{solids}}$);
- $\rho_{sd,bulk}$ is the local bulk density of the sediment (kg/L)
- $v_{sed,acc}$ is the net sediment accumulation rate in water;
- $v_{sed,gross}$ is the gross sedimentation rate from water to sediment (m/s).

$$\rho_{sd,bulk} = 1 - f_{V_{solid,sd}} + f_{V_{solid,sd}} \cdot \rho_{s,sed} \quad \text{Equation 40}$$

- $\rho_{sd,bulk}$ is the local bulk density of the sediment (kg/L)
- $f_{V_{solid,sd}}$ is the volume fraction solids in sediment, with a fixed default of 0.2 (Paterson and Mackay, 1995) (-)
- $\rho_{s,sed}$ is the mineral density of the sediment, with a fixed default of 2.1633 (Fantke et al., 2016) ($\text{kg}_{\text{solids}}/\text{L}_{\text{solids}}$);

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